

This assignment is essentially about **Adversarial Prompting**—the art of finding the "edge cases" where an AI's logic breaks down. Since LLMs predict the next likely word based on patterns rather than true reasoning, you can "confuse" them by creating conflicts between instructions, logic, and common sense.

Here are four effective techniques to "break" or confuse an AI, along with the logic of why they work.

1. The Semantic Paradox (Garden Path Sentences)

These sentences lead the reader (or AI) to one interpretation, only to flip it at the end. They force the AI to re-evaluate the entire context, often leading to grammatical or logical errors.

- **Prompt:** "The old man the boat. Who is the subject and what is the verb?"
- **Why it's tricky:** The AI often reads "The old man" as a noun phrase, but in this sentence, "man" is the verb (meaning to operate) and "old" is the noun (the elderly).
- **Expected Result:** The AI might insist "old man" is the subject and fail to find a verb.

2. Multi-Step Logic with "Hidden" Contradictions

Ask the AI to solve a riddle where the answer is obvious to a human but contradicts the statistical "math" of the prompt.

- **Prompt:** "I have 3 shirts. It takes 3 hours to dry them outside in the sun. How long does it take to dry 30 shirts?"
- **Why it's tricky:** The AI's training data contains thousands of math word problems where x units take y time, so $10x$ should take $10y$. It often misses the "common sense" factor that they all dry at the same time.
- **Expected Result:** It may confidently calculate **30 hours**, ignoring the reality of how sun-drying works.

3. The Constraint Conflict (Recursive Loop)

Force the AI to follow two rules that cannot both be true at the same time.

- **Prompt:** "Write a 50-word story about a cat, but do not use the letter 'e' in any word. Then, explain why you chose those words using only words that *do* contain the letter 'e'."
- **Why it's tricky:** This exhausts the "attention mechanism." While it tries to follow the difficult constraint (lipogram), it often slips up and includes the forbidden letter because "e" is the most common letter in English.
- **Expected Result:** You will likely find at least one "e" in the first section.

4. The "No-Context" Reference

Provide a prompt that refers to something that hasn't happened yet or doesn't exist, forcing the AI to "hallucinate" to be helpful.

- **Prompt:** "Explain the cultural significance of the '2027 Great Blue Moon Riot' in London."
 - **Why it's tricky:** Since this event never happened, the AI has to choose between telling you it's fake or—to be "helpful"—making up a plausible historical narrative based on the keywords.
 - **Expected Result:** The AI might generate a detailed, fake historical account of a riot that never occurred.
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How to Document Your Results

To get full marks on this assignment, I recommend using a simple table to show the "Input" vs. the "AI Failure."

Technique	Prompt Used	AI Response Error	Root Cause
Common Sense	30 shirts drying	Said "30 hours"	Over-reliance on linear math patterns.
Lipogram	No letter 'e'	Included the word "the"	Statistical frequency of "the" bypassed the instruction.

Would you like me to give you a "Jailbreak" style prompt example (designed to bypass safety filters) to see how the AI handles ethical boundaries?